Food Biotechnology Laboratory Course

Sourdough Production and Fermentation Analysis

Laboratory Manual

1 Introduction

Modern food production often uses controlled fermentation to make foods such as bread, dairy products, plant-based foods, and functional foods. Fermentation is not judged only by taste. Instead, scientists measure microbial activity, acidity, and the formation of different chemical compounds to understand how the process is working.

In this laboratory course, sourdough is used as a model system to learn important analytical methods. Even though the ingredients are simple (flour and water), the biological processes behind fermentation are complex and informative. Sourdough is a mixture of flour and water that contains a stable community of microorganisms, mainly lactic acid bacteria (LAB) and yeasts. These microorganisms come from the flour, the environment, or starter cultures.[1] During fermentation, these microbes break down carbohydrates and, to a smaller extent, proteins and fats. This produces lactic acid, acetic acid, ethanol, carbon dioxide (CO_2), and many other metabolites.[2] As organic acids build up, the pH of the dough decreases. This lower pH helps preserve the dough, reduces mold growth, and changes the structure of gluten and starch, which improves texture and shelf life.[3]

Each analytical method in this course reflects a real industry application:

- **pH and TTA measurements** are routinely used in quality control to monitor fermentation progress, product stability, and shelf life.
- **Microbial plating and flow cytometry** are used to assess microbial safety, starter culture performance, and process consistency in fermented foods.
- **Metabolomics (LC-MS)** is applied in research and product development to understand flavor formation, nutritional changes, and metabolic pathways.
- **Multivariate data analysis (PCA, PLS-DA)** is widely used in R&D to compare formulations, optimize fermentation conditions, and support evidence-based product innovation.

The main aim of this course is not to become an expert in sourdough, but to understand how **microbial ecosystems** and **analytical techniques** are used to **design, monitor, and improve fermented food systems**.

2 Learning objectives

After completing this practical course section, you will be able to:

- Prepare and propagate sourdough fermentations.
- Assess total acidity in fermented product (pH, TTA).
- Monitor bacterial and yeast populations using microbiological methods and flow cytometry.
- Perform untargeted metabolomics analysis.
- Scientifically document experimental design, results, and conclusions in a structured report.

3 Experimental Overview and Timeline

The experimental workflow spans ten days (18-27 February 2026) and consists of wet-lab experimentation, dry-lab data analysis, and independent report writing. An overview of the timeline is provided in Table 1.

Experimental timeline

Assisted lab wet-lab session | **Assisted dry lab-session** | **Individual wet-lab work / homework**

Date		Laboratory activities	Independent work	Submission deadlines
Lab Day 1 (t1)	**Wed, 18 Feb 2026**	Sourdough preparation with four flour types; initial sampling	Set up report template; begin literature search IMPORTANT: Include the corresponding hazard (H) and precautionary (P) statements of the chemicals used in the practicum in the Supplementary Material of the report.	
Sourdough Feeding Day (t3)	**Fri, 20 Feb 2026**	Sourdough refreshment and incubation, pH measurements	Writing the introduction section	
Sourdough Feeding Day (t6)	**Mon, 23 Feb 2026**	Sourdough refreshment and incubation, pH measurements	Writing the methods section	Optional: Submit the introduction and methods sections by **Tuesday, 24 February 2026 on `Moodle` by 18:00**.
Lab Day 2 (t8)	**Wed, 25 Feb 2026**	Final refreshment and sampling (t8); baking of sourdough breads		
Lab Day 3	**Fri, 27 Feb 2026**	Data analysis, interpretation.	Completion of the final report	Submit report by Thursday, **2 April 2026 on `Moodle` by 18:00.**

Disclaimer:
The indicated start and end times are approximate and may vary depending on laboratory progress, group size, and technical issues.

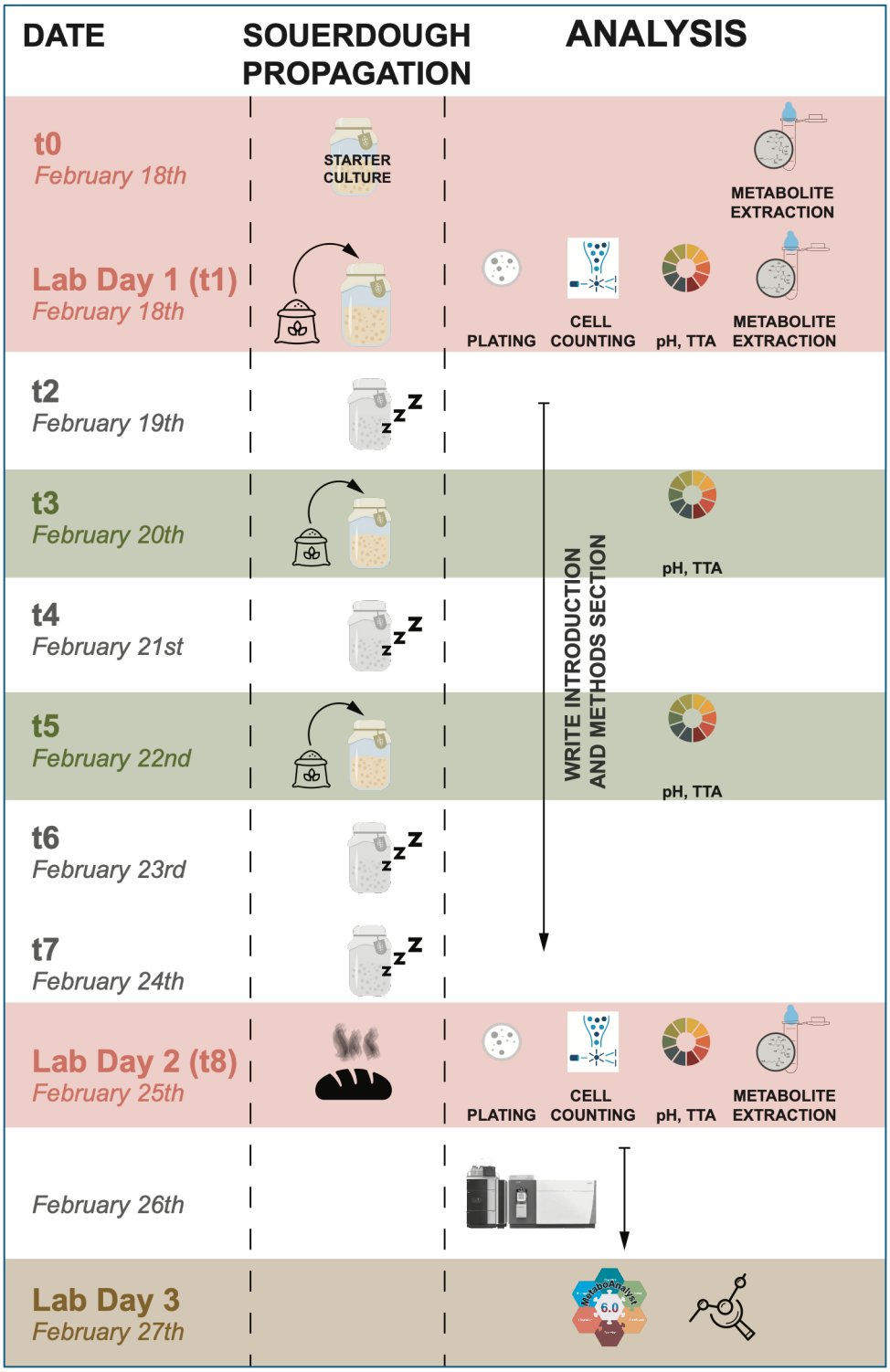


Figure 1. Experimental schedule of the sourdough production and fermentation practical.

4 Safety and Good Laboratory Practice

All laboratory work must always follow general safety rules. Appropriate personal protective equipment, including lab coat, gloves, and safety glasses, must be worn, and biological samples and chemicals should be handled according to instructions. For any questions regarding the procedures, please refer to the lab assistants, who are happy to help.

Biosafety

Sourdough intended for baking must remain in the food-grade laboratory. Samples intended for analytical measurements must be transferred into Falcon tubes and transported to the designated analytical laboratory under BSL-1 conditions. Only microorganisms classified as BSL-1 (non-pathogenic) are expected.

Chemical safety:

Key chemical hazards used in this practicum:

- Sodium hydroxide (NaOH): corrosive base
- Acetonitrile (ACN): flammable and toxic organic solvent
- Propidium iodide (PI): irritant and suspected mutagen.

* Sodium hydroxide solution 0.1 M Safety Information

Hazard statements		
	H290	May be corrosive to metals.
Precautionary statements		
Prevention	P234	Keep only in original packaging
	P270	Wear protective gloves/eye protection
Response	P314	Get medical attention/advice if you feel unwell
	P390	Absorb spillage to prevent material damage

Safety data sheet from Carl ROTH, revision date 10.10.2024, accessed: LC 02.02.2026.

* Propidium iodide Safety Information

Hazard statements		
	H315	Causes skin irritation.
	H319	Causes serious eye irritation
	H335	May cause respiratory irritation
	H341	Suspected of causing genetic defects
Precautionary statements		
Prevention	P261	Avoid breathing dust
	P270	Wear protective gloves/eye protection
Response	P314	Get medical attention/advice if you feel unwell
	P302+P352	**Skin** IF ON SKIN: Wash with plenty of water.
	P305+P351+P338	**Eyes** IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
	P301+P312 P330	**Ingestion** IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell Rinse mouth

Safety data sheet from Carl ROTH, revision date 17.09.2024, accessed: LC 02.02.2026.

* Acetonitrile Safety Information

Hazard statements		
	H225	Highly flammable liquid and vapor
	H302+H312+H332	Harmful if swallowed, in contact with skin or if inhaled
	H319	Causes serious eye irritation
Precautionary statements		
Prevention	P264	Wash face, hands and any exposed skin thoroughly after handling
	P270	Do not eat, drink or smoke when using this product
	P280	Wear protective gloves/protective clothing/eye protection/face protection
	P271	Use only outdoors or in a well-ventilated area
	P261	Do not breathe dust/fume/gas/mist/vapors/spray
	P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
	P233	Keep container tightly closed
	P240	Ground and bond container and receiving equipment
	P241	Use explosion-proof electrical/ventilating/lighting equipment
	P243	Take action to prevent static discharges
	P242	Use non-sparking tools
Response	P314	Get medical attention/advice if you feel unwell
		Inhalation
	P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing
	P312	Call a POISON CENTER or doctor if you feel unwell
		Skin
	P312	Call a POISON CENTER or doctor if you feel unwell
	P362+P364	Take off contaminated clothing and wash before reuse
	P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
		Eyes
	P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
	P337+P313	If eye irritation persists: Get medical advice/attention
		Ingestion
	P301+P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
	P330	Rinse mouth
Fire	P370+P378	In case of fire: Use CO_2, dry chemical, or foam to extinguish
Storage	P403+P235	Store in a well-ventilated place. Keep cool
Disposal	P501	Dispose of contents/container to an approved waste disposal plant

Safety data sheet from Thermo Fisher for Acetonitrile, revision date 18.12.2025, accessed: LC 14.01.2026.

Safety Note:

The NaCl-peptone solution used for dilutions is **not considered hazardous** under normal laboratory handling. Peptone is a protein extract used to grow microbes and may cause mild irritation if it contacts eyes or skin.

Waste disposal:

Waste must be disposed of in the designated containers:

- Biological waste (sourdough residues, plates): biohazard waste
- Organic solvents (ACN): halogen-free organic waste
- NaOH solutions: aqueous basic waste
- Sharps and glass: designated containers

5 Raw Materials and Starter Cultures

5.1 Flours Used

Different cereal and pseudo-cereal flours are used to compare fermentation behavior and microbial development. Each group uses two different flour types for comparison:

Group A/B: Wheat flour (Weizenmehl) and Spelt flour (Dinkelmehl)

Group C/D: Spelt flour (Dinkelmehl) and Buckwheat flour (Buchweizenmehl)

Group E/F: Buckwheat flour (Buchweizenmehl) and Maize flour (Maismehl)

Group G/H: Maize flour (Maismehl) and Wheat flour (Weizenmehl)

Disclaimer:
Alternatively, use self-provided flour for sourdough propagation.

5.2 Starter Culture

A sourdough starter from wheat is used to ensure reproducibility and comparability between experimental groups.

Notes

PROTOCOL DAY 1

6 Sourdough Preparation and Fermentation

6.1 Initial Sourdough Preparation

Each group prepares two sourdoughs with different flour types. These will be compared during the course.

6.1.1 Materials

- Sourdough starter culture (t0) in 2 L jar
- Your flour
- Spoons
- Bottled water
- Glass jars

6.1.2 Procedure

1. Weigh the clean jar and write it down. This will be important to feed the sourdough during the next days.
2. Add 50 g of sourdough starter to each of the two jars.
3. Add 30 g of flour and 30 g of water.
4. Mix thoroughly until a homogeneous dough is obtained, take a sample (t1).
5. Incubate at room temperature for 48 to 72 hours before the next refreshment.

6.2 Sampling at timepoint t1

For all analyses, approximately 20 g of sourdough are required. The samples are weighed into Falcon tubes and transferred to the analytical laboratory.

For the initial sampling at t1 these analyses should be done:

- Physicochemical analysis: pH and TTA measurements
- Microbial analysis: plating, flow cytometry
- Metabolite analysis: metabolite extraction

Important:
Each group must consistently use the same flour type and labeling scheme throughout the entire experiment to ensure correct data interpretation.

PROTOCOL DAY 2-3

6.3 Sourdough Refreshment (t 1-3)

Safety note:
Sourdough refreshment must be done in the food-grade laboratory. The sourdoughs are refreshed 3 times, with 2-3 days in between refreshments.

6.3.1 Procedure

1. Take a sample for pH measurement. For this, fill about 15 g of sourdough into a Falcon tube.
2. Keep 50 g of your sourdough and use the rest for analysis, or discard if there is no analysis that day.
3. Add 25 g of your flour and 25 mL of mineral water to the jar.
4. Mix thoroughly using a sterile spoon until a homogeneous dough is obtained.
5. Incubate at room temperature for 48 to 72 hours before the next refreshment.
6. Take the sample to the analytical lab (leave the rest of the sourdough in the food-grade laboratory) and measure pH. Enter the pH value in the results file on **Moodle** (sourdough_results_sheet.xlsx)

PROTOCOL DAY 4

6.4 Final Sampling after 1 week

For the final sampling, 20 g of sourdough is required. Take 20 g of sourdough and use the rest for the baking of sourdough breads.

For the final sampling, the following analyses should be performed:

- Physicochemical analysis: pH and TTA measurements
- Microbial analysis: plating, flow cytometry
- Metabolite analysis: metabolite extraction

6.5 Baking Sourdough Breads

Each experimental group prepares a small loaf using one of their sourdough cultures. Don't use your starter, if it has mold on top.

6.5.1 Ingredients

- 50 g sourdough starter
- 400 g flour (half your flour, half wheat flour)
- 320 mL water
- 2 teaspoons salt

6.5.2 Procedure

1. Mix the water, sourdough starter, and dissolved salt. Add the flour and knead for 10 minutes. Cover and allow to ferment at room temperature for 1.5 hours.
2. Fold the dough. Repeat this twice after 30 minutes and 60 minutes. Allow it to ferment for an additional 1.5 hours.
3. Preheat the oven to 230 °C. Place the dough in a floured baking pan and score the surface. Bake for 30 minutes. Remove the foil and bake for an additional 15 minutes, or until golden brown.

7 Wet-Lab Analysis

In this experiment, each sourdough sample weighs **20 g**. You will use it as follows:

- **pH and TTA:** Take **10 g** from the sample. You can use the same portion for both pH and TTA measurements.
- **Microbial analysis:** Take **5 g** from the sample. First, prepare the serial dilutions for **plating**. Then use the **10^{-2} dilution** from that preparation for **flow cytometry**.
- **Metabolomics:** Take **105 mg (0.105 g)**. Be careful when weighing this small amount on the balance.

After these steps, about **5 g** of the original sample will remain.

7.1 Microbial Analysis: Plating and Colony Counting

To characterize the microbial composition of sourdough, a culture-based approach using selective agar plates is applied to support the growth of specific microbial groups while inhibiting others.[4] In this experiment, the focus is on lactic acid bacteria and yeasts, the two dominant microbial groups responsible for sourdough fermentation.

Serial dilution of the sourdough sample ensures that countable numbers of colonies are obtained on agar plates. After incubation, visible colonies originate from individual viable cells or small cell aggregates present in the original sample. By counting colonies at appropriate dilution levels and accounting for dilution factors and sample mass, microbial abundances can be calculated and expressed as colony-forming units per gram of sourdough (CFU/g).[5] This quantitative approach enables comparison of bacterial and yeast populations across different flour types and fermentation stages and provides insight into microbial dynamics during sourdough propagation.

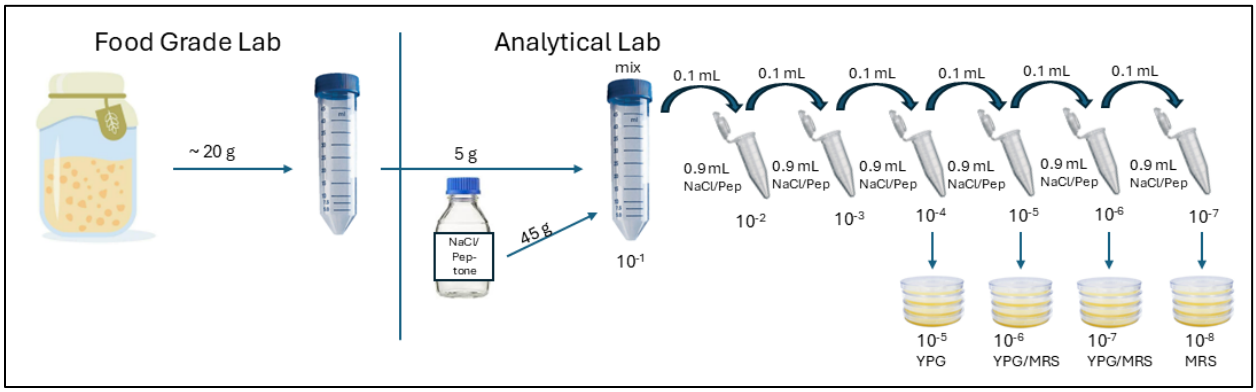


Figure 2. Plate count analysis of sourdough samples.

7.1.1 Materials

- Balance
- Plastic spoon (sterile)
- 5 g of sourdough sample
- 50 mL Falcon tube (sterile)
- NaCl-peptone-solution (0.85% NaCl, 1% peptone), 45 mL and 6 x 900 µL for aliquots
- 100 µL pipette and tips
- L-shaped culture spreaders (sterile)
- Agar-media plates: 2 different media, 3 plates per medium
  - 3x Man-Rogosa-Sharpe (MRS) medium: selective growth medium for lactic acid bacteria (LAB)
  - 3x Yeast extract peptone glycerol (YPG) medium: growth medium supporting yeasts

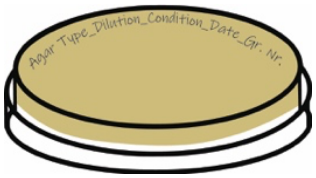
7.1.2 Procedure

1. Using a sterile spoon, weigh 5 g of sourdough into a 50 mL Falcon tube.
2. Add 45 g (yes grams!) of sterile NaCl-peptone solution to the same Falcon tube to obtain a 10^{-1} initial dilution.

   Note:
   Pour approximately 43-44 g of peptone solution from the shot glass into a Falcon tube and use a pipette to bring the total to 45 g.

3. Homogenize the suspension thoroughly by vortexing. If the dough adheres to the tube walls, place the tube horizontally on the vortex mixer and gently rotate it to facilitate detachment.
4. Prepare the first serial dilution by aseptically transferring 0.1 mL of the homogenized suspension (10^{-1}) into 0.9 mL of sterile NaCl-peptone solution in a sterile 1.5 mL microcentrifuge tube. Vortex thoroughly to obtain a 10^{-2} dilution.
5. Continue the decimal serial dilution by transferring 0.1 mL from each dilution tube into the next tube containing 0.9 mL of sterile NaCl-peptone solution, mixing thoroughly by vortexing at each step before the next transfer. This procedure yields successive dilutions of 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, and 10^{-7}.
6. Label all agar plates clearly with:

   (1) agar type (MRS or YPG)
   (2) final dilution (YPG: 10^{-5}, 10^{-6}, 10^{-7}; MRS: 10^{-6}, 10^{-7}, 10^{-8})
   (3) condition
   (4) date
   (5) group number.



Important:

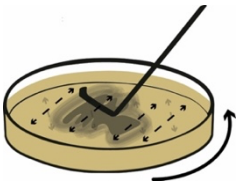
- From this step onward, perform all plating steps close to a Bunsen burner or in a sterile working area to minimize contamination.
- During pipetting and spreading, work from the highest dilution (10^{-8}) to the lowest dilution (10^{-5}).

7. For:
   - 3 MRS agar plates: Pipette 0.1 mL of the 10^{-5}, 10^{-6}, 10^{-7} dilutions onto the center of the labeled MRS plates to obtain final plate dilutions of 10^{-6}, 10^{-7}, 10^{-8}.
   - 3 YPG agar plates: Pipette 0.1 mL of the 10^{-4}, 10^{-5}, 10^{-6} dilutions onto the center of the labeled YPG plates to obtain final plate dilutions of 10^{-5}, 10^{-6}, 10^{-7}.

Important:

- Use a **single** pipette tip for **each condition**: **pipette strictly** from the highest dilution (10-8) to the lowest (10-5) to minimize carryover contamination.
- Plating 0.1 mL introduces an additional 10^{-1} dilution factor.

8. Spread the inoculum evenly over the entire agar surface using an L-shaped spreader under aseptic conditions using one sterile spreader per condition:
   - Spread the plates strictly in order from the highest dilution (10^{-8}) to the lowest dilution (10^{-5}) to minimize carryover contamination.



9. Allow the inoculum to absorb into the agar surface with the lids closed, then submit the plates for incubation.
10. Incubate the plates inverted at 30 °C for 48 hours.
11. After incubation, count and evaluate the colonies. Only plates containing 10-300 colonies should be used for accurate enumeration.

Important:
If the colony count on the highest dilution plate exceeds 300 colonies, count one quarter of the plate and multiply the result by four, provided the colonies are evenly distributed.

12. If possible, count colonies from at least two suitable dilutions (preferably all three) and record the results in the provided results file on **Moodle** (sourdough_results_sheet.xlsx)

For the Report:
Calculate microbial counts as colony-forming units per gram (CFU/g) of sourdough. Ensure that all dilution factors (initial homogenization, serial dilutions, and plating volume) are correctly included in the calculation.

7.2 Physicochemical Analysis: pH and TTA Measurement

Sourdough fermentation is driven by microbial metabolism, primarily by lactic acid bacteria and yeasts, which convert carbohydrates into organic acids such as lactic and acetic acid. The accumulation of these acids lowers the pH, affecting dough structure, microbial stability, and fermentation kinetics. Monitoring acidification is therefore essential for evaluating fermentation progress, typically using pH and total titratable acidity (TTA).[6]

pH measures the concentration of free hydrogen ions in a solution, reflecting only the acids that are dissociated, and is expressed as a unit-less value. TTA, on the other hand, quantifies the total acidity, including both dissociated and undissociated acids. It is measured by titrating a known amount of sourdough with a standard base (commonly 0.1 M NaOH) until all acids are neutralized. This method accounts for both free hydrogen ions and those bound in weak acids that react with the base. TTA is expressed as milliliters of 0.1 M NaOH per gram of dough.[7]

By combining pH and TTA measurements, a comprehensive picture of acidification can be obtained, as pH reflects free hydrogen ions while TTA reflects total acid production.

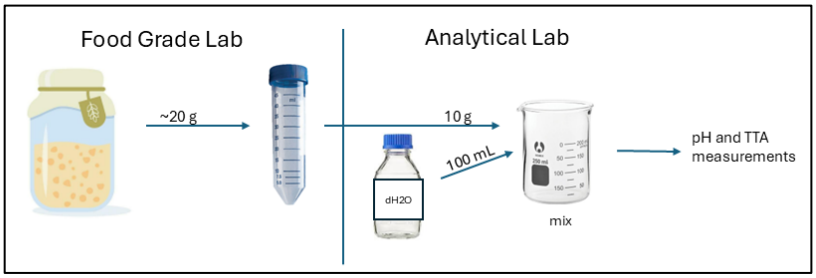


Figure 3. pH and TTA analysis of sourdough samples.

7.2.1 Materials

- Balance
- 200 mL beaker
- Magnetic stir bar and stirrer
- Deionized water
- 0.1 M NaOH* solution (*see “Safety Information”)
- Protective glasses
- 10 g sourdough
- 1000 µL pipette and tips
- pH meter

7.2.2 Procedure

Before starting, calibrate the pH meter with pH 4.01, 7.00 and 9.21.

Safety note:
Always make sure to wear protective glasses when working with NaOH.

pH measurement

1. Place the beaker on the balance.
2. Weigh 10 g of your sourdough directly in the beaker.
3. Add 100 mL of deionized water (deionized tap) and a magnetic stir bar.
4. Place the beaker on the stirrer and let the mixture stir until it is fully homogenized.
5. Determine the pH value with the pH-meter.
6. Note down the measured value before continuing with the titration.

TTA

1. Place the NaOH bottle on the balance besides the stirrer. Tare it to be able to monitor how much NaOH you will take out of the bottle.
2. With the mixture still stirring, gradually add 0.1 M NaOH to your sourdough suspension by **SLOWLY PIPETTING** the NaOH solution from the 0.1 M bottle to the sourdough mix in the beaker. Keep an eye on the increasing pH.
3. Continue adding more NaOH solution until you reach pH 8.5.
   Note: Once pH 8.0 is reached, add 0.1 M NaOH with increased caution to avoid overshooting the target pH of 8.5. Note down the volume/weight of 0.1 M NaOH you had to add to reach pH 8.5.
   Add both the pH and TTA values in the Excel sheet on **`Moodle`**.
   (`sourdough_results_sheet.xlsx`).

7.3 Flow cytometry analysis

Flow cytometry enables rapid single-cell analysis of complex microbial communities in liquid suspension. In this practicum, flow cytometry is used to quantify total bacterial and yeast cells, distinguish live and dead cells, and estimate bacteria-to-yeast ratios in sourdough samples. Compared to traditional plating methods, flow cytometry is expected to detect higher cell counts, as it includes both culturable and non-culturable cells that do not grow on agar. During measurement, individual particles are transported in a fluid stream through a laser beam and characterized based on light scattering and fluorescence signals. Forward scatter provides information on particle size, while side scatter reflects internal complexity and granularity.[8] Prior to analysis, cells are fluorescently stained to enable population discrimination. SYBR Green permeates all cells and binds to DNA, allowing detection of total microbial cells, whereas propidium iodide penetrates only cells with compromised membranes and therefore selectively stains dead cells.[9]

Optional:

- Introduction to flow cytometry: https://youtu.be/ZSF6QFofUts?si=5s7Udc55OFsovBKp (accessed 13.02.2026).
- For a more detailed explanation of flow cytometry and the stepwise identification of specific cell populations through gating, refer to **Annex Section 11.1**.

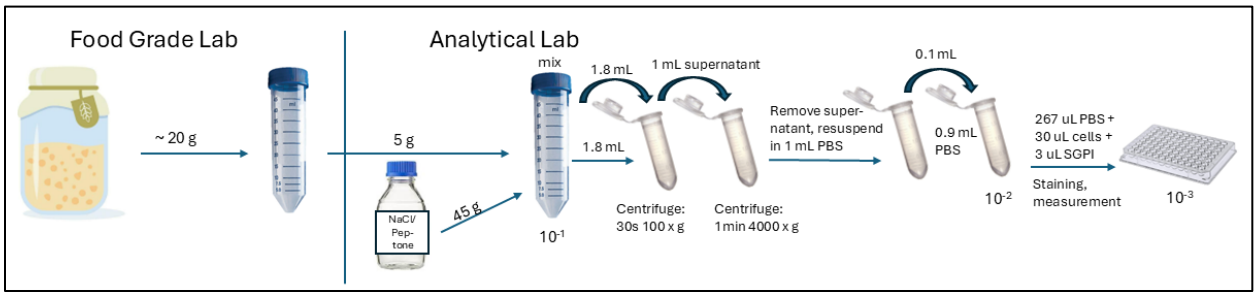


Figure 4. Flow cytometry analysis of sourdough samples.

7.3.1 Materials

- Sourdough, 1:10 dilution in peptone solution (previously used for plating)
- PBS 1× (filtered)
- 2 mL Eppendorf tubes
- SYBR Green (SG) and propidium iodide* (PI) working solutions (*see “Safety Information”)
- 96-well plate

7.3.2 Procedure

Safety note:
Wear protective gloves and eye protection.

Microbial cell preparation and washing

All samples are prepared in **triplicate**.

Label:
`YYYYMMDD_<sample_type>_timepoint_<group_nr>`

Where:

- `YYYYMMDD` → Date of extraction (e.g., `20250225`)
- `<sample_type>` → Flour type (e.g., `spelt`)
- `timepoint` → Timepoint of sampling (e.g., `t0, t1, t2`...)
- `group_nr` → Sample number (e.g., `1, 7`...)

Examples:

- Sample: `20250225_spelt_t0_1`

1. Transfer 1.8 mL of the 1:10 dilution into a 2 mL Eppendorf tube prepared for the plating experiment.
2. Centrifuge for 30 s at 100 × g to sediment large flour particles while keeping microbial cells in suspension.
3. Transfer 1 mL of the supernatant containing bacterial and fungal cells into a new 2 mL Eppendorf tube.
4. Centrifuge for 1 min at 4000 × g to pellet the cells (yeast and bacteria).
5. Carefully discard the supernatant, leaving the cell pellet intact.
6. Resuspend the pellet in 1 mL of filtered PBS. The suspension remains a 1:10 dilution.
7. Perform an additional 1:10 dilution by transferring 100 µL of the cell suspension into 900 µL of filtered PBS in a 2 mL Eppendorf tube. This final suspension is used for fluorescent staining.

Staining and plate preparation

Stain information:

- **SYBR Green (SG):** permeates all cells and stains total DNA, allowing detection of all microbial cells.
- **Propidium iodide (PI):** penetrates only cells with compromised membranes and therefore selectively stains dead cells.

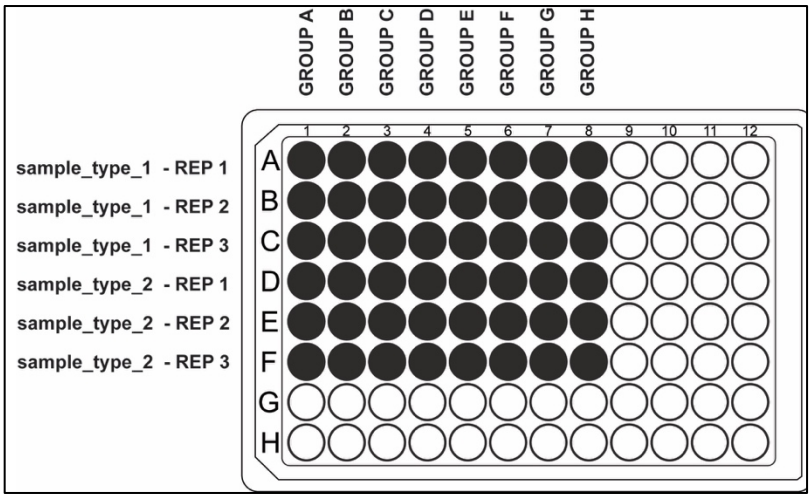


Figure 5. Arrangement 96-well plate for flow analysis.

Table 1. Preparation of the staining mixture.

Sample	Step 1: PBS (µL)	Step 2: Cell Suspension (µL)	Step 3: Dye (µL)	Final Conditions / Notes
SGPI	267	30	3 (SGPI)	1:10 dilution, 1× SG, 4 µM PI

8. Samples are transferred to a 96-well plate according to Figure 6.
9. Prepare the staining mixtures according to the table 1.
10. Gently tap the 96-well plate to ensure thorough mixing.
11. Incubate the plate for 15-30 min in the dark (e.g., in a cupboard).
12. After incubation, samples are ready for measurement on the CytoFLEX flow cytometer.

7.4 Metabolite Extraction and LC-MS Analysis

Metabolomics is the large-scale study of small molecules that provides direct insight into microbial metabolism, growth, and interactions, and thus offers a functional snapshot of the biochemical state of sourdough.[10] In metabolomics, the focus is on low-molecular-weight compounds, typically with mass-to-charge ratios (m/z) below 2000, which can be analyzed using liquid chromatography-mass spectrometry (LC-MS).[11] The workflow begins with metabolite extraction from the sourdough matrix, followed by the removal of insoluble debris and proteins to obtain a clarified extract. Chromatographic separation is then performed using high-performance liquid chromatography (HPLC) with a column selected according to the physicochemical properties of the target metabolites. In this practical course, a hydrophilic interaction liquid chromatography (HILIC) column is employed to preferentially retain polar compounds, while non-polar molecules elute more rapidly.[12] Following chromatographic separation, the metabolites are ionized and introduced into the mass spectrometer. Under high-vacuum conditions, the resulting ions are directed along defined trajectories and detected according to their mass-to-charge ratios (m/z). Structural information is subsequently obtained from their fragmentation spectra, enabling putative metabolite annotation.[13] The final output is a feature table containing the detected ions, their respective peak areas, and, where possible, putative metabolite annotations. These features can subsequently be compared across experimental conditions to assess how microbial activity and fermentation dynamics influence the biochemical profile of the sourdough.

Optional:

- Overview of how a mass spectrometer operates: https://youtu.be/K1VSYjuw6os?si=LmouYzzD8TulfNmO (accessed 13.02.2026).
- For a more detailed explanation of the data-dependent acquisition (DDA) method in mass spectrometry, refer to **Annex Section 11.2.**

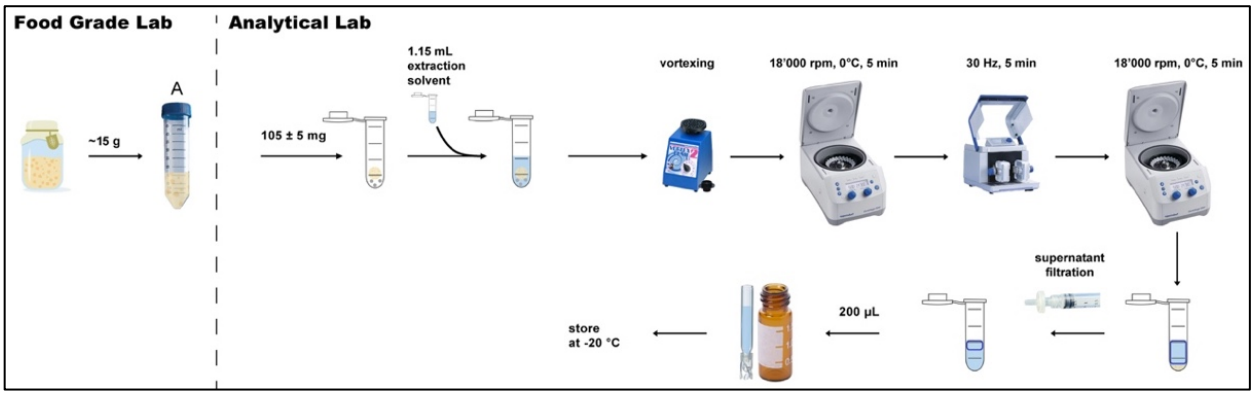


Figure 6. Metabolite extraction of sourdough samples.

7.4.1 Materials

- Extraction solvent: Ice-cold 80% (v/v) acetonitrile* (ACN) in water, LC-MS grade (Fisher Scientific) (*see “Safety Information”) supplemented with 500 nM CUDA (molecular mass 340.5) as extraction internal standard (IS).
- Balance: Mettler Toledo XP205 DeltaRange, readability d = 0.01 mg / 0.1 mg
- Vortex Genie 2™, Bender & Hobein AG, Zürich, Switzerland
- TissueLyser III, Qiagen
- Centrifuge 5425R, Eppendorf, 0 °C, 15’000 rpm
- DNA LoBind Eppendorf Tubes, 2 mL, Safe-Lock, PCR clean, Eppendorf SE, Germany
- Disposable Luer-Slip Syringe (Polypropylene), 2 mL, 100 pcs, HPLC certified, Charge 8000033949, BGB, Germany
- Syringe Filter P7N, 4 mm PTFE (hydrophilic), 0.22 µm, pack of 200, S/N 281044188, BGB, China
- SureSTART™ CAP AVCS, white silicone, pack of 100, Thermo Scientific, made in Germany
- 1.5 mL Short Thread Vials, transparent, wide opening, 32×11.6 mm, pack of 100
- 250 µL Conical Glass Inserts, total volume 340 µL, 31×6 mm, pack of 100, BGB, Germany
- 3 mm stainless steel beads

7.4.2 Procedure

Safety note:
Make sure to wear protective glasses during this procedure.

1. Weigh 105 ± 5 mg of the sample into a 2 mL Eppendorf tube containing three 3 mm stainless steel beads.

   Important:
   - Be careful **not to spill the beads**.
   - Pay close attention to the **decimal on the balance**: 100 mg = 0.1 g.

2. Add 1.15 mL of extraction solvent to each tube.
3. Vortex briefly.
4. Centrifuge at 15’000 rpm, 0 °C for 5 min.
5. Homogenize using TissueLyser III at 30 Hz for 5 min.
6. Centrifuge again at 15’000 rpm, 0 °C for 5 min.
7. Filter the supernatant using a Luer-slip syringe with a 4 mm PTFE filter into a labeled 2 mL Eppendorf tube.

Filtration Step:

1. Take a Luer-slip syringe and attach the 4 mm PTFE filter to the tip.
2. Remove the plunger from the syringe.
3. Carefully pipette your solution (about 1 mL) into the syringe barrel, making sure not to touch the pellet.
4. Re-insert the plunger and slowly push the solution through the filter.
5. Collect the filtered supernatant into a labeled 2 mL Eppendorf tube.

⚠️ **Tip**: Push the plunger slowly and steadily to avoid splashing or spilling.

8. Bring the 2 mL Eppendorf tube containing the pellet and the three metal beads to the supervisors.
9. Pipette 200 µL of the filtered extract into **labeled** LC-MS vial.

Important:

Make sure your LC-MS vial is labeled as indicated!

10. Pipette another 200 µL of the filtered extract into the QC 15 mL falcon tube (there is only one QC tube for all groups for all conditions and all timepoints).

Information:

The QC (Quality Control) sample is used to:

- Prime the mass spectrometer (show the machine how typical sample should look)
- To check that signal intensities stay consistent throughout the run

Label:

`YYYYMMDD_<sample_type>_timepoint_<group_nr>`

Where:

- `YYYYMMDD` → Date of extraction (e.g., `20250225`)
- `<sample_type>` → Flour type (e.g., `spelt`) or `MB` for method blanks
- `timepoint` → Timepoint of sampling (e.g., `t0, t1, t2`…)
- `group_nr` → Sample number or replicate (e.g., `1, 7`…)

Examples:

- Sample: `20250225_spelt_t0_1`

11. Store labeled Eppendorf tubes and LC-MS vials at -20 °C until analysis.

8 Data Interpretation and Reporting

Scientific writing is a key learning outcome of this practical course. The final report should demonstrate that the experimental design and data can be interpreted and discussed in a scientific manner. It is expected that the report will compare microbial, physicochemical, culture-independent, and metabolomic data across the different sourdough fermentations,

Code:

The code used to generate the figures should be deposited in the Supplementary Material.

For the Report:

If you are unsure about the structure of a scientific report, refer to **Annex Section 10.3**.

8.1 Physicochemical Analysis of Sourdough

The evolution of pH and titratable acidity (TTA) during sourdough propagation at room temperature was monitored over 7 days. Two different flour types were used per group to renew the sourdough at each refreshment step. Measurements were taken at timepoints 1, 3, 5, and 8. TTA is measured as milliliters of 0.1 M NaOH required to titrate 10 g of sourdough to pH 8.5. Do not forget to normalize to 1 g of dough. Data should be interpreted at least for the two conditions prepared by your group. Additional conditions can also be analyzed if desired. One recommended option is to prepare a plot showing pH and TTA over time to visualize the progression of sourdough acidification.

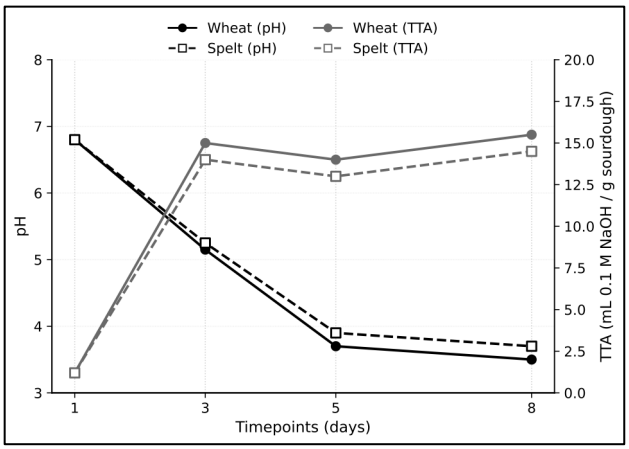


Figure 7. Physiochemical analysis of sourdough conditions using estimated data.

8.2 Bacterial and yeast cells

8.2.1 Plating

For the plating-based analysis, you have plates for both bacteria (MRS medium) and yeast (YPG medium). From these plates:

1. Count colonies for each dilution.
2. Calculate the number of bacterial and yeast cells per gram of sourdough (CFU/g), considering the dilution series and the initial amount of sourdough used.
3. Report mean and standard deviation (SD) if multiple replicates were counted.

!	It should be noted that CFU values represent only culturable viable cells and may underestimate the total microbial abundance compared to flow cytometry.

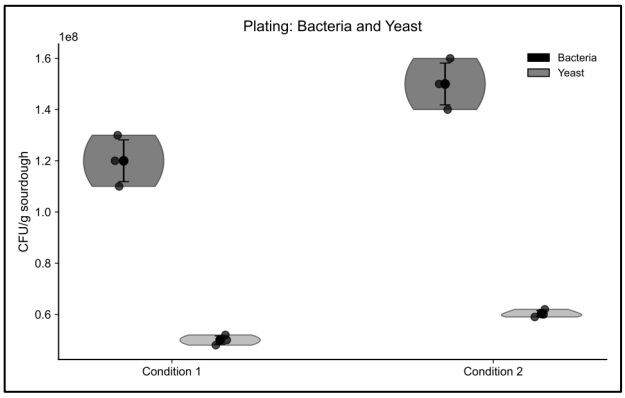


Figure 8. Microbial Counts in Sourdough by Plating.

8.2.2 Flow Cytometry

For the flow cytometry data:

1. Generate a plot showing live bacteria and live yeast counts per sample. Include mean and SD for the triplicates of each flour type.
2. Use the information about the measurement parameters:
    a. Measurement time per sample (1 min)
    b. Flow rate (30 µL/min)
3. Calculate the total volume analyzed (flow rate × measurement time).
4. Using the dilution factors applied during sample preparation and the initial amount of sourdough, calculate the number of cells per gram of sourdough.

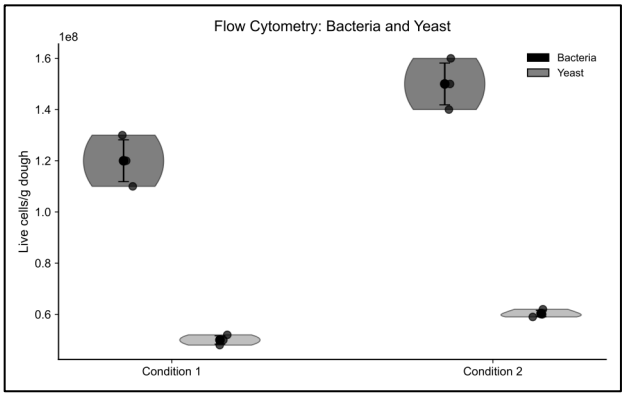


Figure 9. Microbial Counts in Sourdough by Flow Cytometry.

8.2.3 Comparison and Interpretation

- Compare the live bacterial and yeast counts obtained by flow cytometry with the counts obtained by plating.
- Discuss whether the numbers correspond.
- If discrepancies are observed, consider possible explanations.

8.3 Untargeted Metabolomics - Lab Day 3

The LC-MS data were converted from the proprietary *.raw* format to *.mzML* using **`MSConvert 3`** and subsequently processed with **`MZmine 4.8.30`**. Metabolite features were annotated using **`SIRIUS 6.3.3`**. Statistical analysis was conducted in a Python environment using the **`Google Colab notebook`** provided on **`Moodle`**. The analysis employs a two-factor experimental design accounting for both time point and flour type to support interpretation of the metabolomics data.

During the session, you will:

1. Data import and inspection: Load metadata and feature table and perform an initial quality check,
2. Univariate analysis
   - Generate volcano plots to identify metabolites that differ significantly between the two sourdough types at the same time point.
3. Preprocessing of metabolomics data
   - Apply appropriate filtering, normalization, and transformation procedures.
4. Multivariate analysis
   - PCA (Principal Component Analysis)
     - Visualize global metabolic variation across all conditions and time points.
   - PLS-DA (Partial Least Squares Discriminant Analysis)
     - Identify the top 15 metabolites contributing most to group separation, comparing the selected condition at the final time point to the other conditions at the same time point.
5. Optional: Pathway enrichment analysis (ORA)
     - Map significant metabolites to biochemical pathways to identify key biological processes influenced by sourdough type and fermentation time.

9 References and Further Reading

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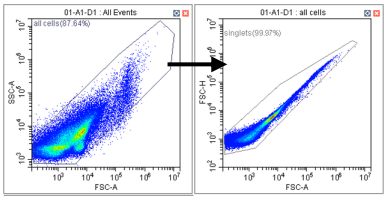
10 Annex

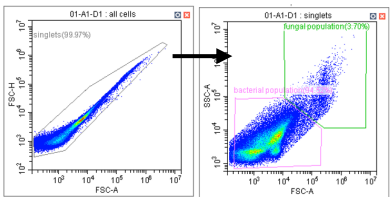
10.1 Flow Cytometry Gating Strategy for Sourdough Microbes

Flow cytometry data analysis requires **gating** to identify the populations of interest and remove artifacts. Gating involves drawing boundaries on plots of different signals so that only relevant cells are analyzed. In this experiment, **FSC-A (forward scatter)** reflects cell size, while **SSC-A (side scatter)** indicates internal complexity, allowing debris and stacked cells to be excluded and bacterial and yeast populations to be separated. **SYBR Green (B525-A)** stains DNA in all cells, identifying the total microbial population, and **propidium iodide (PI, Y690-A)** stains only cells with compromised membranes, marking dead cells. Here is a clear stepwise explanation of the gating used in this experiment:

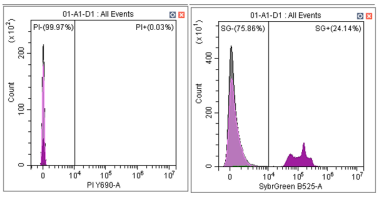
1. **Remove debris and aggregates (doublets)**
   Exclude small non-cellular particles (debris) and stacked or clumped cells that give inaccurate signals.



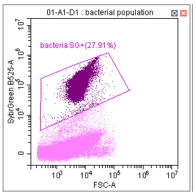
2. **Separate bacterial and fungal (yeast) populations**
   Different microbial types have distinct sizes and internal complexity; this step distinguishes bacteria from yeast based on their scatter properties.



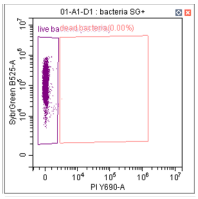
3. **PI (Y690-A) vs Count and SYBR Green (B525-A) vs Count**
   Examining positive and negative boundaries for PI and SYBR Green before applying them to the gated populations.



4. **Identify total bacterial cells**
   SYBR Green stains DNA in all cells, allowing detection of total bacterial population.

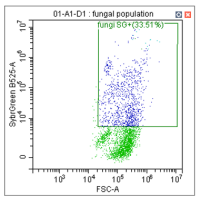


5. **Determine bacterial viability**
   Propidium iodide (PI) penetrates only dead cells. SG+ PI− cells are live bacteria, SG+ PI+ are dead.



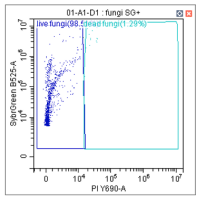
6. Identify total fungal (yeast) cells

SYBR Green stains DNA in all cells, allowing detection of total fungal population.



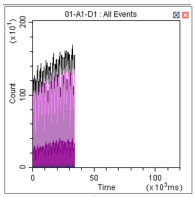
7. Determine fungal (yeast) viability

Propidium iodide (PI) penetrates only dead cells. SG+ PI− cells are live fungi, SG+ PI+ are dead.



8. Monitor count over time

Ensures consistent sample flow and identifies potential anomalies during data acquisition.



10.2 Mass Spectrometry Data-Dependent Acquisition (DDA)

In this practical course, a data-dependent acquisition (DDA) method was performed using an Orbitrap mass spectrometer. The instrument operates in repeated cycles that alternate between MS1 and MS2 scans.

During the MS1 scan, all precursor ions within the selected m/z range are detected, and their signal intensities provide quantitative information. Based on this scan, the instrument automatically selects the top N most intense precursor ions in real time. These ions are then isolated and fragmented in the MS2 (MS/MS) scan to generate fragment spectra.

Each MS/MS spectrum represents the fragmentation pattern of a specific precursor ion. These characteristic fragment patterns are subsequently used for compound identification by comparison with spectral libraries or *in silico* fragmentation databases.

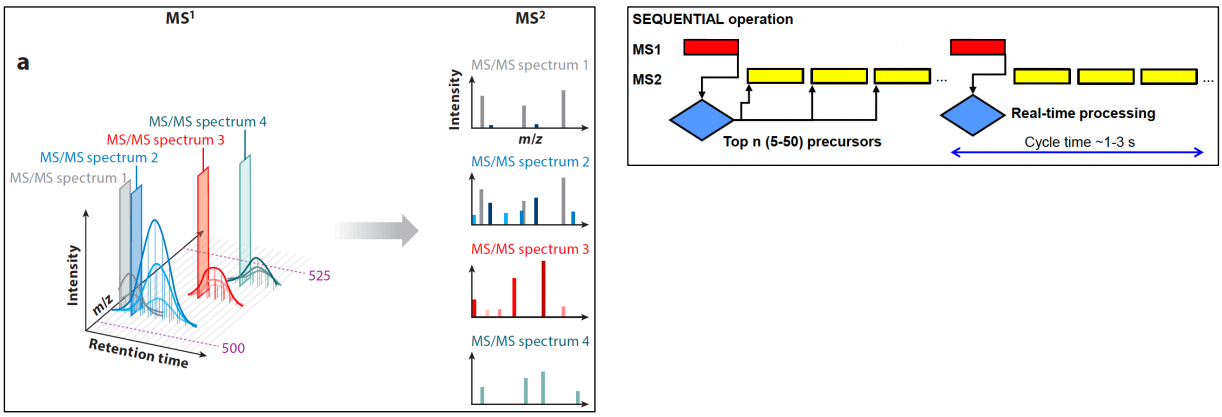


Figure 10. Data-Dependent Acquisition (DDA) Workflow. **Left:** Schematic of MS1 and MS2 scans. The MS1 chromatogram shows signal intensity over retention time, while the MS2 spectrum displays fragment ion intensity versus m/z. **Right:** Sequential operation in an Orbitrap instrument. An MS1 scan is first acquired, followed by MS2 scans of the top *N* most intense precursor ions selected in real time for fragmentation. Figure adapted from the Advanced Proteomics Block Course 2025, “Proteomics Using Data-Dependent Acquisition Strategies,” Alexander Leitner.

10.3 Scientific Report Structure

A complete scientific report must include (in accordance with Lecture 389-6000-00L: *Academic Writing and Publishing in Molecular Biology*).

10.3.1 Title

- Concise and informative
- Clearly reflects the main experimental question

10.3.2 Abstract

The abstract summarizes the entire study and should be understandable to readers outside your immediate field.

Recommended structure:

- **1-2 sentences:** General introduction to sourdough fermentation, understandable to any scientist
- **2-3 sentences:** More detailed background relevant to food fermentation and microbiology
- **1 sentence:** Clear statement of the general problem or research question addressed in this study
- **1 sentence:** Summary of the main results (use formulations such as *"Here we show that …"*)
- **2-3 sentences:** Explanation of how the results extend or challenge previous knowledge
- **1-2 sentences:** Broader scientific or practical context of the findings
- **2-3 sentences:** Outlook or broader perspective, accessible to scientists from any discipline

Reasoning: The abstract is often the only part many readers read. It must therefore be self-contained, precise, and informative. Use around 150-300 words.

10.3.3 Keywords

- 4-6 relevant keywords
  Examples: *sourdough fermentation, flour type, lactic acid bacteria, LC-MS metabolomics*

10.3.4 Introduction

The introduction should guide the reader from general background knowledge to the specific research question.

Structure (three major parts):

1. General background
   - Introduce sourdough fermentation and its relevance (food quality, tradition, microbiology)
2. Current state of knowledge and knowledge gap
   - Summarize what is already known about sourdough microbiology and flour composition
   - Identify what is *not* known or insufficiently studied
   - Clearly state the knowledge gap

3. Research objective and hypothesis
   - Present the aim of this experiment
   - Formulate the main research question and, if appropriate, hypotheses

Reasoning: A strong introduction motivates the study and prepares the reader to understand the results.

10.3.5 Materials and Methods

The aim of this section is to provide all information required to understand and reproduce the experiments.

- Describe materials (flours, starter culture, instruments, software)
- Describe experimental procedures clearly and logically
- Use past tense and passive voice where appropriate

Important: Another scientist should be able to reproduce your experiment based on this section.

10.3.6 Results

The results section presents **observations and measurements only**, without interpretation.

General guidelines:

- Begin by briefly referring to the research question and explaining the experimental approach
- Use **clear subheadings** with declarative titles (what was found, not what was done)
- Organize results into repeated logical units: *Question → Experiment → Result*
- Use figures and tables to illustrate key findings
  - Refer to them in the text and highlight their main message
  - Do not repeat figure legends in the text

Do	Do not
- Present results objectively and concisely, use past tense - Use numbers and statistics where appropriate Example: *"The pH of wheat sourdough was significantly lower than that of maize sourdough ($p < 0.05$, t-test)."*	- Interpret results or provide explanations (discussion) - Include extensive background information (introduction) - Present raw data or intermediate calculations (supplementary material)

10.3.7 Discussion

The discussion explains **what the results mean** and how they answer the research question.

- Interpret your findings in the context of existing literature
- Explain similarities or differences compared to previous studies
- Discuss limitations and potential sources of error
- Highlight the relevance and implications of your findings

Style tips:

- Mostly present tense and active voice
- Clear logical structure with subheadings
- Avoid repeating results; instead, refer to specific figures or data when needed

10.3.8 Conclusion

The conclusion briefly summarizes the main findings and their significance.

- Answer the research question clearly
- Highlight the most important outcome of the study
- Indicate practical implications or directions for future research

10.3.9 Literature

- Cite all sources used in the report
- Use a reference manager (e.g. EndNote or Zotero)

10.3.10 Supplementary Material

Supplementary material contains additional information that supports the main text but is too detailed to include in the main report.

Typical content:

- Large tables
- Extended datasets
- Additional figures
- Detailed calculations

Structure:

- Use clear headings or bullet points
- Refer to supplementary items explicitly in the main text (e.g. *"see Supplementary Table S1"*)
  Example: *Supplementary Table S1: Complete LC-MS feature table for all sourdough samples*

Reasoning: Supplementary material improves clarity of the main text while ensuring transparency and reproducibility.